

Q1. Basic File Operation:

```
_/m/'
Last login: Sat Sep 26 03:20:14 2026 from 13.233.177.4
[ec2-user@Nikita ~]$ sudo -i
[root@Nikita ~]# mkdir LinuxExam
[root@Nikita ~]# ls
LinuxExam
[root@Nikita ~]# cd LinuxExam
[root@Nikita LinuxExam]# touch student.txt
[root@Nikita LinuxExam]# touch course.txt
[root@Nikita LinuxExam]# result.txt
-bash: result.txt: command not found
[root@Nikita LinuxExam]# touch result.txt
[root@Nikita LinuxExam]# pwd
/root/LinuxExam
[root@Nikita LinuxExam]# ls -l
total 0
-rw-r--r--. 1 root root 0 Sep 26 03:29 course.txt
-rw-r--r--. 1 root root 0 Sep 26 03:29 result.txt
-rw-r--r--. 1 root root 0 Sep 26 03:29 student.txt
[root@Nikita LinuxExam]#
```

Q.2 File Management:

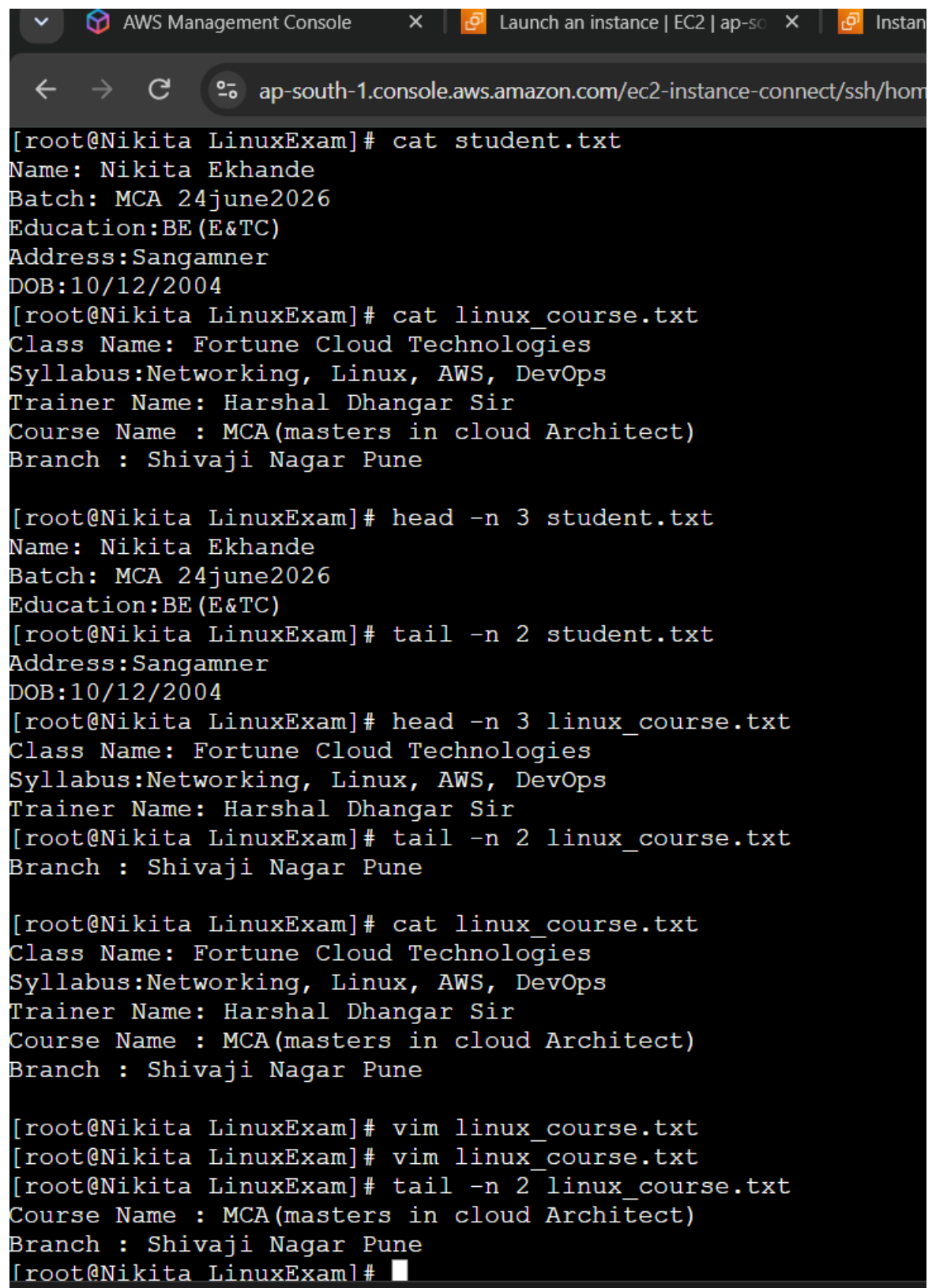
```
[root@Nikita LinuxExam]# mv student.txt student_backup.txt
[root@Nikita LinuxExam]# ls
course.txt  result.txt  student_backup.txt
[root@Nikita LinuxExam]# mv course.txt linux_course.txt
[root@Nikita LinuxExam]# ls
linux_course.txt  result.txt  student_backup.txt
[root@Nikita LinuxExam]# rm result.txt
rm: remove regular empty file 'result.txt'? y
[root@Nikita LinuxExam]# ls
linux_course.txt  student_backup.txt
[root@Nikita LinuxExam]# mkdir Documents
[root@Nikita LinuxExam]# mkdir Backups
[root@Nikita LinuxExam]# mkdir Scripts
[root@Nikita LinuxExam]# ls
Backups  Documents  Scripts  linux_course.txt  student_backup.txt
[root@Nikita LinuxExam]# mv student_backup.txt Backups
[root@Nikita LinuxExam]# tree
-bash: tree: command not found
[root@Nikita LinuxExam]# ls -R
.:
Backups  Documents  Scripts  linux_course.txt

./Backups:
student_backup.txt

./Documents:

./Scripts:
[root@Nikita LinuxExam]#
```

Q3.File Content Operation:



The screenshot shows the AWS Management Console interface with a terminal window open. The terminal displays a series of commands and their outputs for file operations on a Linux instance named 'Nikita LinuxExam'.

```
[root@Nikita LinuxExam]# cat student.txt
Name: Nikita Ekhande
Batch: MCA 24june2026
Education:BE (E&TC)
Address:Sangamner
DOB:10/12/2004
[root@Nikita LinuxExam]# cat linux_course.txt
Class Name: Fortune Cloud Technologies
Syllabus:Networking, Linux, AWS, DevOps
Trainer Name: Harshal Dhangar Sir
Course Name : MCA(masters in cloud Architect)
Branch : Shivaji Nagar Pune

[root@Nikita LinuxExam]# head -n 3 student.txt
Name: Nikita Ekhande
Batch: MCA 24june2026
Education:BE (E&TC)
[root@Nikita LinuxExam]# tail -n 2 student.txt
Address:Sangamner
DOB:10/12/2004
[root@Nikita LinuxExam]# head -n 3 linux_course.txt
Class Name: Fortune Cloud Technologies
Syllabus:Networking, Linux, AWS, DevOps
Trainer Name: Harshal Dhangar Sir
[root@Nikita LinuxExam]# tail -n 2 linux_course.txt
Course Name : MCA(masters in cloud Architect)
Branch : Shivaji Nagar Pune

[root@Nikita LinuxExam]# cat linux_course.txt
Class Name: Fortune Cloud Technologies
Syllabus:Networking, Linux, AWS, DevOps
Trainer Name: Harshal Dhangar Sir
Course Name : MCA(masters in cloud Architect)
Branch : Shivaji Nagar Pune

[root@Nikita LinuxExam]# vim linux_course.txt
[root@Nikita LinuxExam]# vim linux_course.txt
[root@Nikita LinuxExam]# tail -n 2 linux_course.txt
Course Name : MCA(masters in cloud Architect)
Branch : Shivaji Nagar Pune
[root@Nikita LinuxExam]#
```

Q4. User Management:

```
aws | [Alt+S] As
[root@Nikita LinuxExam]# passwd student01
Changing password for user student01.
New password:
Retype new password:
passwd: all authentication tokens updated successfully.

tcpdump:x:72:72:::/sbin/nologin
ec2-user:x:1000:1000:EC2 Default User:/home/ec2-user:/bin/bash
student01:x:1001:1001::/home/student01:/bin/bash
[root@Nikita LinuxExam]# su student01
[student01@Nikita LinuxExam]$ username

[student01@Nikita LinuxExam]$ id student01
uid=1001(student01) gid=1001(student01) groups=1001(student01)
[student01@Nikita LinuxExam]$
```

Q5. Group Management:

```
ec2-user:x:1000:
student01:x:1001:
linuxbatch:x:1002:student01
```

```
[root@Nikita ~]# groupadd linuxbatch
[root@Nikita ~]# gpasswd -a student01 linuxbatch
Adding user student01 to group linuxbatch
[root@Nikita ~]# cat /etc/group
root:x:0:
```

```
student01:x:1001:1001::/home/student01:/bin/bash
[root@Nikita ~]# id student01
uid=1001(student01) gid=1001(student01) groups=1001(student01),1002(linuxbatch)
[root@Nikita ~]#
```

```
[root@Nikita ~]# useradd -m student02
[root@Nikita ~]# gpasswd -a student02 linuxbatch
Adding user student02 to group linuxbatch
[root@Nikita ~]# cat /etc/passwd
```

```
ec2-user:x:1000:
student01:x:1001:
linuxbatch:x:1002:student01,student02
student02:x:1003:
[root@Nikita ~]#
```

Q6. File Permission:

```
Last login: Sat Sep 26 03:23:56 2026 from 13.233.177.5
[ec2-user@Nikita ~]$ sudo -i
[root@Nikita ~]# touch project.txt
[root@Nikita ~]# chmod 751 project.txt
[root@Nikita ~]# ls -l
total 0
drwxr-xr-x. 5 root root 96 Sep 26 04:05 LinuxExam
-rwxr-x--x. 1 root root  0 Sep 26 05:00 project.txt
[root@Nikita ~]# chmod 640 project.txt
[root@Nikita ~]# chown student01 project.txt
[root@Nikita ~]# ls -l
total 0
drwxr-xr-x. 5 root      root 96 Sep 26 04:05 LinuxExam
-rw-r-----. 1 student01 root  0 Sep 26 05:00 project.txt
[root@Nikita ~]# chgrp linuxbatch project.txt
[root@Nikita ~]# ls -l
total 0
drwxr-xr-x. 5 root      root      96 Sep 26 04:05 LinuxExam
-rw-r-----. 1 student01 linuxbatch  0 Sep 26 05:00 project.txt
[root@Nikita ~]#
```

Q7 Permission Challenge:

```
[root@Nikita ~]# mkdir -p LinuxExam/ public/ private/shared/
[root@Nikita ~]# ls -l LinuxExam
total 8
drwxr-xr-x. 2 root root 32 Sep 26 03:36 Backups
drwxr-xr-x. 2 root root  6 Sep 26 03:36 Documents
drwxr-xr-x. 2 root root  6 Sep 26 03:36 Scripts
-rw-r--r--. 1 root root 187 Sep 26 04:05 linux_course.txt
-rw-r--r--. 1 root root  95 Sep 26 04:02 student.txt
[root@Nikita ~]# ls -R LinuxExam
LinuxExam:
Backups Documents Scripts linux_course.txt student.txt

LinuxExam/Backups:
student_backup.txt

LinuxExam/Documents:

LinuxExam/Scripts:
[root@Nikita ~]# chmod 755 LinuxExam/public
```

```
[root@Nikita ~]# chmod 755 public
[root@Nikita ~]# ls -ld
dr-xr-x---. 6 root root 184 Sep 26 05:06 .
[root@Nikita ~]# chmod 700 private
[root@Nikita ~]# ls -ld
dr-xr-x---. 6 root root 184 Sep 26 05:06 .
[root@Nikita ~]# chmod 750 private
[root@Nikita ~]# ls -ld
dr-xr-x---. 6 root root 184 Sep 26 05:06 .
[root@Nikita ~]#
```

Q8. Package Management:

```
Last login: Sat Sep 26 04:59:55 2026 from 13.233.177.5
[ec2-user@Nikita ~]$ sudo -i
[root@Nikita ~]# yum upgrade
Last metadata expiration check: 0:20:28 ago on Sat Sep 26 05:14:04 2026.
Dependencies resolved.
Nothing to do.
Complete!
[root@Nikita ~]# yum install httpd -y
Last metadata expiration check: 0:20:41 ago on Sat Sep 26 05:14:04 2026.
Package httpd-2.4.68-1.amzn2023.0.1.x86_64 is already installed.
Dependencies resolved.
Nothing to do.
Complete!
```

```
[root@Nikita ~]# httpd -v
Server version: Apache/2.4.68 (Amazon Linux)
Server built: Jun 8 2026 00:00:00
```

```
[root@Nikita ~]# systemctl start httpd
[root@Nikita ~]# systemctl status httpd
● httpd.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/httpd.service; disabled; preset: disabled)
   Active: active (running) since Sat 2026-09-26 05:41:55 UTC; 58s ago
     Docs: man:httpd.service(8)
  Main PID: 32109 (httpd)
    Status: "Total requests: 0; Idle/Busy workers 100/0; Requests/sec: 0; Bytes served/sec: 0 B/sec"
    Tasks: 177 (limit: 1059)
   Memory: 13.7M
      CPU: 135ms
   CGroup: /system.slice/httpd.service
           └─32109 /usr/sbin/httpd -DFOREGROUND
           └─32132 /usr/sbin/httpd -DFOREGROUND
           └─32134 /usr/sbin/httpd -DFOREGROUND
           └─32159 /usr/sbin/httpd -DFOREGROUND
           └─32172 /usr/sbin/httpd -DFOREGROUND
```

```
Sep 26 05:41:55 Nikita systemd[1]: Starting httpd.service - The Apache HTTP Server...
Sep 26 05:41:55 Nikita httpd[32109]: AH00558: httpd: Could not reliably determine the server's fully qualified domain name, using fe80::8ff:d1ff:feaa:a9%
Sep 26 05:41:55 Nikita systemd[1]: Started httpd.service - The Apache HTTP Server.
Sep 26 05:41:55 Nikita httpd[32109]: Server configured, listening on: port 80

[root@Nikita ~]# systemctl enable httpd
Created symlink /etc/systemd/system/multi-user.target.wants/httpd.service → /usr/lib/systemd/system/httpd.service.
[root@Nikita ~]#
```

Q9. Apache Webserver Configuration:

```
bash: cd: /var/www/html/: No such file or directory
[root@Nikita ~]# cd /var/www/html/
[root@Nikita html]# ls
[root@Nikita html]# vim index.html
[root@Nikita html]# vim index.html
[root@Nikita html]# vim index.html
[root@Nikita html]# vim index.html
[root@Nikita html]# vim index.html
[root@Nikita html]# systemctl restart httpd
-bash: systemctl: command not found
[root@Nikita html]# systemctl restart httpd
[root@Nikita html]# curl -i http://localhost
HTTP/1.1 200 OK
Date: Sun, 27 Sep 2026 14:22:23 GMT
Server: Apache/2.4.68 (Amazon Linux)
Last-Modified: Sun, 27 Sep 2026 14:19:32 GMT
ETag: "52-65c77a36d90ee"
Accept-Ranges: bytes
Content-Length: 82
Content-Type: text/html; charset=UTF-8

<h1> Nikita Ekhande </h1>
<h2> Roll No: 1 </h2>
<p> Linux Course Examination </p>
[root@Nikita html]#
```

← → ↻ ⚠ Not secure 13.127.114.232

Nikita Ekhande

Roll No: 1

Linux Course Examination

Q10. Process Management:

```
R-xfs-buf/nvme0n1p1]
root    1059  0.0  0.0      0  0 ?      I<   14:10  0:00 [kworker/R-xfs-conv/nvme0n1p1]
root    1060  0.0  0.0      0  0 ?      I<   14:10  0:00 [kworker/R-xfs-reclaim/nvme0n1p1]
root    1061  0.0  0.0      0  0 ?      I<   14:10  0:00 [kworker/R-xfs-blockgc/nvme0n1p1]
root    1062  0.0  0.0      0  0 ?      I<   14:10  0:00 [kworker/R-xfs-inodegc/nvme0n1p1]
root    1063  0.0  0.0      0  0 ?      I<   14:10  0:00 [kworker/R-xfs-log/nvme0n1p1]
root    1064  0.0  0.0      0  0 ?      I<   14:10  0:00 [kworker/R-xfs-cil/nvme0n1p1]
root    1065  0.0  0.0      0  0 ?      S    14:10  0:00 [xfsaild/nvme0n1p1]
root    1112  0.0  1.7  54264 16276 ?    Ss   14:10  0:00 /usr/lib/systemd/systemd-journald
root    1545  0.0  1.2 32884 11720 ?    Ss   14:10  0:00 /usr/lib/systemd/systemd-udevd
systemd+ 1570  0.0  1.6 23508 15632 ?    Ss   14:10  0:00 /usr/lib/systemd/systemd-resolved
root    1572  0.0  0.2 21936 2680 ?     $csl 14:10  0:00 /sbin/auditd
root    1576  0.0  0.0      0  0 ?      I<   14:10  0:00 [kworker/R-rpciod]
root    1577  0.0  0.0      0  0 ?      I<   14:10  0:00 [kworker/R-xprtiod]
root    1587  0.0  0.0      0  0 ?      I<   14:10  0:00 [kworker/R-ena]
root    1696  0.0  0.7 17092 7316 ?     Ss   14:10  0:00 /usr/bin/systemd-inhibit --what=handle-suspend-key:handle-hibernate-key --who=ec2-hibi
root    1699  0.0  0.3 81436 3120 ?     Ssl   14:10  0:00 /usr/sbin/irqbalance --foreground
libstor+ 1700  0.0  0.2 2788 2032 ?      Ss   14:10  0:00 /usr/bin/lsmc -d
root    1703  0.0  0.8 17596 8380 ?     Ss   14:10  0:00 /usr/lib/systemd/systemd-homed
root    1704  0.0  1.1 19436 10800 ?    Ss   14:10  0:00 /usr/lib/systemd/systemd-logind
dbus    1705  0.0  0.4 8532 4180 ?      Ss   14:10  0:00 /usr/bin/dbus-broker-launch --scope system --audit
systemd+ 1708  0.0  1.1 237680 10388 ?    Ss   14:10  0:00 /usr/lib/systemd/systemd-networkd
dbus    1714  0.0  0.3 5396 3048 ?      S    14:10  0:00 dbus-broker --log 4 --controller 9 --machine-id ec285ff22355f1ed01acb3dd2213ee87 --max
root    1732  0.0  0.2 2700 1872 ?      S    14:10  0:00 /usr/sbin/acpid -f
root    1737  0.0  0.4 282732 3920 ?     Ssl   14:10  0:00 /usr/sbin/gssproxy -D
root    2046  0.0  2.1 1242044 19832 ?   Ssl   14:10  0:00 /usr/bin/amazon-ssm-agent
root    2051  0.0  0.7 9600 6804 ?      Ss   14:10  0:00 sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups
root    2057  0.0  0.2 4792 2664 ?      Ss   14:10  0:00 /usr/sbin/atd -f
root    2058  0.0  0.2 221376 1948 tty1  Sst+ 14:10  0:00 /sbin/agetty -o -p -- \u --nclcar - linux
root    2059  0.0  0.2 221420 2004 ttyS0 Sst+ 14:10  0:00 /sbin/agetty -o -p -- \u --keep-baud 115200,57600,38400,9600 - vt220
chrony  2079  0.0  0.4 87152 3964 ?      S    14:10  0:00 /usr/sbin/chronyd -F 2
root    2119  0.0  1.1 15852 10676 ?    Ss   14:10  0:00 sshd-session: ec2-user [priv]
root    2404  0.0  0.7 17092 7264 ?      Ss   14:10  0:00 /usr/lib/systemd/systemd-userdbd
ec2-user 2409  0.0  1.5 22796 14348 ?    Ss   14:10  0:00 /usr/lib/systemd/systemd --user
ec2-user 2411  0.0  0.9 109000 8664 ?     S    14:10  0:00 (sd-pam)
ec2-user 2418  0.0  0.7 16104 7264 ?      R    14:10  0:00 sshd-session: ec2-user@pts/0
```

```
aws Search [Alt+S] Asia Pacific (Mumbai) v
ec2-user 2419  0.0  0.5 224096 5008 pts/0 Ss 14:10 0:00 -bash
root    2774  0.0  0.0      0  0 ?      I    14:15  0:00 [kworker/1:2-cgroup_free]
root    2895  0.0  0.0      0  0 ?      I    14:18  0:00 [kworker/0:1-events]
root    2963  0.0  1.2 19348 11648 ?    Ss   14:20  0:00 /usr/sbin/httpd -DFOREGROUND
apache  2964  0.0  0.5 19004 4832 ?      S    14:20  0:00 /usr/sbin/httpd -DFOREGROUND
top - 14:24:19 up 14 min, 2 users, load average: 0.00, 0.00, 0.00
Tasks: 114 total, 1 running, 113 sleeping, 0 stopped, 0 zombie
%Cpu(s): 3.1 us, 0.0 sy, 0.0 ni, 96.9 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
MiB Mem : 913.2 total, 483.8 free, 169.5 used, 260.0 buff/cache
MiB Swap: 0.0 total, 0.0 free, 0.0 used, 611.5 avail Mem

  PID USER      PR  NI   VIRT    RES    SHR   S %CPU  %MEM    TIME+  COMMAND
    1 root        20   0 173200 17628 10948 S   0.0   1.9   0:00.99 systemd
    2 root        20   0      0      0      0 S   0.0   0.0   0:00.00 kthreadd
    3 root        20   0      0      0      0 S   0.0   0.0   0:00.00 pool_workqueue_release
    4 root       -20   0      0      0      0 I   0.0   0.0   0:00.00 kworker/R-rcu_gp
    5 root       -20   0      0      0      0 I   0.0   0.0   0:00.00 kworker/R-sync_wq
    6 root       -20   0      0      0      0 I   0.0   0.0   0:00.00 kworker/R-kvfree_rcu_reclaim
    7 root       -20   0      0      0      0 I   0.0   0.0   0:00.00 kworker/R-slub_flushwq
    8 root       -20   0      0      0      0 I   0.0   0.0   0:00.00 kworker/R-netns
    9 root        20   0      0      0      0 I   0.0   0.0   0:00.04 kworker/0:0-mm_percpu_wq
   10 root       -20   0      0      0      0 I   0.0   0.0   0:00.00 kworker/0:0H-events_highpri
   13 root       -20   0      0      0      0 I   0.0   0.0   0:00.00 kworker/R-mm_percpu_wq
   14 root        20   0      0      0      0 S   0.0   0.0   0:00.02 ksoftirqd/0
   15 root        20   0      0      0      0 I   0.0   0.0   0:00.02 rcu_top - 14:24:22 up 14 min, 2 users, load average: 0.00, 0.00, 0.00

%Cpu(s): 0.0 us, 0.0 sy, 0.0 ni,100.0 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st

   17 root        20   0      0      0      0 S   0.0   0.0   0:00.00 rcu_

3204 apache      20  0 1545728 8128  4580 S   0.3   0.9   0:00.07 httpd
3468 root        20  0 224072 3536  2844 R   0.3   0.4   0:00.01 top
    1 root        20  0 173200 17628 10948 S   0.0   1.9   0:00.99 systemd
    2 root        20  0      0      0      0 S   0.0   0.0   0:00.00 kthreadd
    3 root        20  0      0      0      0 S   0.0   0.0   0:00.00 pool_workqueue_release
```

```
    5 root       -20   0      0      0      0 I   0.0   0.0   0:00.00 kworker/R-sync_wq
    6 root       -20   0      0      0      0 I   0.0   0.0   0:00.00 kworker/R-kvfree_rcu_reclaim
    7 root       -20   0      0      0      0 I   0.0   0.0   0:00.00 kworker/R-slub_flushwq
    8 root       -20   0      0      0      0 I   0.0   0.0   0:00.00 kworker/R-netns
    9 root        20   0      0      0      0 I   0.0   0.0   0:00.04 kworker/0:0-mm_percpu_wq
   10 root       -20   0      0      0      0 I   0.0   0.0   0:00.00 kworker/0:0H-events_highpri
   13 root       -20   0      0      0      0 I   0.0   0.0   0:00.00 kworker/R-mm_percpu_wq
   14 root        20   0      0      0      0 S   0.0   0.0   0:00.02 ksoftirqd/0
   15 root        20   0      0      0      0 I   0.0   0.0   0:00.02 rcu_top - 14:24:25 up 14 min, 2 users, load average: 0.00, 0.00, 0.00

top - 14:24:25 up 14 min, 2 users, load average: 0.00, 0.00, 0.00
xp_par_gp_kthread_worker/0
%Cpu(s): 0.0 us, 0.2 sy, 0.0 ni, 99.8 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
xp_gp_kthread_worker

tion/0
26 root        20  0      0      0      0 I   0.3   0.0   0:00.25 kworker/u8:1-flush-259:0
    1 root        20  0 173200 17628 10948 S   0.0   1.9   0:00.99 systemd

i-0468f668416315de (Linux_exam)
```

```
ation/0
26 root      20    0      0      0      0 I  0.3  0.0  0:00.25 kworker/u8:1-flush-259:0
1 root      20    0 173200 17628 10948 S  0.0  1.9  0:00.99 systemd
2 root      20    0      0      0      0 S  0.0  0.0  0:00.00 kthreadd
3 root      20    0      0      0      0 S  0.0  0.0  0:00.00 pool_workqueue_release
4 root      0 -20    0      0      0 I  0.0  0.0  0:00.00 kworker/R-rcu_gp
5 root      0 -20    0      0      0 I  0.0  0.0  0:00.00 kworker/R-sync_wq
6 root      0 -20    0      0      0 I  0.0  0.0  0:00.00 kworker/R-kvfree_rcu_reclaim
7 root      0 -20    0      0      0 I  0.0  0.0  0:00.00 kworker/R-slub_flushwq
8 root      0 -20    0      0      0 I  0.0  0.0  0:00.00 kworker/R-netns
9 root      20    0      0      0      0 I  0.0  0.0  0:00.04 kworker/0:0-mm_percpu_wq
10 root     0 -20    0      0      0 I  0.0  0.0  0:00.00 kworker/0:0H-events_highpri
13 root     0 -20    0      0      0 I  0.0  0.0  0:00.00 kworker/R-mm_percpu_wq
14 root     20    0      0      0      0 S  0.0  0.0  0:00.02 ksoftop - 14:24:34 up 14 min,  2 users,  load average: 0.00, 0.00, 0.00

%Cpu(s):  0.0 us,  0.2 sy,  0.0 ni, 99.7 id,  0.0 wa,  0.0 hi,  0.2 si,  0.0 st

16 root      20    0      0      0      0 S  0.0  0.0  0:00.00 rcu_

1065 root    20    0      0      0      0 S  0.3  0.0  0:00.21 xfsaild/nvme0n1p1
2965 apache  20    0 1545728 8188  4576 S  0.3  0.9  0:00.07 httpd
2986 apache  20    0 1709632 7460  3796 S  0.3  0.8  0:00.07 httpd
1 root      20    0 173200 17628 10948 S  0.0  1.9  0:00.99 systemd
2 root      20    0      0      0      0 S  0.0  0.0  0:00.00 kthreadd
3 root      20    0      0      0      0 S  0.0  0.0  0:00.00 pool_workqueue_release
4 root      0 -20    0      0      0 I  0.0  0.0  0:00.00 kworker/R-rcu_gp
5 root      0 -20    0      0      0 I  0.0  0.0  0:00.00 kworker/R-sync_wq
6 root      0 -20    0      0      0 I  0.0  0.0  0:00.00 kworker/R-kvfree_rcu_reclaim
7 root      0 -20    0      0      0 I  0.0  0.0  0:00.00 kworker/R-slub_flushwq
8 root      0 -20    0      0      0 I  0.0  0.0  0:00.00 kworker/R-netns
9 root      20    0      0      0      0 I  0.0  0.0  0:00.04 kworker/0:0-mm_percpu_wq
10 root     0 -20    0      0      0 I  0.0  0.0  0:00.00 kworker/0:0H-events_highpri
13 root     0 -20    0      0      0 I  0.0  0.0  0:00.00 kworker/R-mm_percpu_wq
14 root     20    0      0      0      0 S  0.0  0.0  0:00.02 ksoftop - 14:24:37 up 14 min,  2 users,  load average: 0.00, 0.00, 0.00

%Cpu(s):  0.0 us,  0.0 sy,  0.0 ni,100.0 id,  0.0 wa,  0.0 hi,  0.0 si,  0.0 st
```

```
14 root      20    0      0      0      0 S  0.0  0.0  0:00.02 ksof

1 root      20    0 173200 17628 10948 S  0.0  1.9  0:00.99 systemd
2 root      20    0      0      0      0 S  0.0  0.0  0:00.00 kthreadd
3 root      20    0      0      0      0 S  0.0  0.0  0:00.00 pool_workqueue_release
4 root      0 -20    0      0      0 I  0.0  0.0  0:00.00 kworker/R-rcu_gp
5 root      0 -20    0      0      0 I  0.0  0.0  0:00.00 kworker/R-sync_wq
6 root      0 -20    0      0      0 I  0.0  0.0  0:00.00 kworker/R-kvfree_rcu_reclaim
7 root      0 -20    0      0      0 I  0.0  0.0  0:00.00 kworker/R-slub_flushwq
8 root      0 -20    0      0      0 I  0.0  0.0  0:00.00 kworker/R-netns
9 root      20    0      0      0      0 I  0.0  0.0  0:00.04 kworker/0:0-mm_percpu_wq
10 root     0 -20    0      0      0 I  0.0  0.0  0:00.00 kworker/0:0H-events_highpri
13 root     0 -20    0      0      0 I  0.0  0.0  0:00.00 kworker/R-mm_percpu_wq
14 root     20    0      0      0      0 S  0.0  0.0  0:00.02 ksoftirqd/0
15 root     20    0      0      0      0 I  0.0  0.0  0:00.02 rcu_top - 14:24:40 up 14 min,  2 users,  load average: 0.00, 0.00, 0.00

%Cpu(s):  0.0 us,  0.2 sy,  0.0 ni, 99.8 id,  0.0 wa,  0.0 hi,  0.0 si,  0.0 to - 14:24:43 up 14 min,  2 users,  load average: 0.00, 0.00, 0.00

%Cpu(s):  0.0 us,  0.0 sy,  0.0 ni, 99.8 id,  0.0 wa,  0.0 hi,  0.0 si,  0.2 st

18 root      rt    0      0      0      0 S  0.0  0.0  0:00.00 migr

1737 root    20    0 282732  3920  2844 S  0.3  0.4  0:00.01 gssproxy
3468 root    20    0 224072  3536  2844 R  0.3  0.4  0:00.02 top
1 root      20    0 173200 17628 10948 S  0.0  1.9  0:00.99 systemd
2 root      20    0      0      0      0 S  0.0  0.0  0:00.00 kthreadd
3 root      20    0      0      0      0 S  0.0  0.0  0:00.00 pool_workqueue_release
4 root      0 -20    0      0      0 I  0.0  0.0  0:00.00 kworker/R-rcu_gp
5 root      0 -20    0      0      0 I  0.0  0.0  0:00.00 kworker/R-sync_wq
6 root      0 -20    0      0      0 I  0.0  0.0  0:00.00 kworker/R-kvfree_rcu_reclaim
7 root      0 -20    0      0      0 I  0.0  0.0  0:00.00 kworker/R-slub_flushwq
8 root      0 -20    0      0      0 I  0.0  0.0  0:00.00 kworker/R-netns
9 root      20    0      0      0      0 I  0.0  0.0  0:00.04 kworker/0:0-mm_percpu_wq
```

```
10 root      0 -20    0      0      0 I  0.0  0.0  0:00.00 kworker/0:0H-events_highpri
13 root      0 -20    0      0      0 I  0.0  0.0  0:00.00 kworker/R-mm_percpu_wq
14 root      20    0      0      0      0 S  0.0  0.0  0:00.02 ksoftirqd/0
15 root      20    0      0      0      0 I  0.0  0.0  0:00.02 rcu_preempt
16 root      20    0      0      0      0 S  0.0  0.0  0:00.00 rcu_exp_par gp kthread_worker/0
Change delay from 3.0 to
18 root      rt    0      0      0      0 S  0.0  0.0  0:00.00 migration/0
19 root      20    0      0      0      0 S  0.0  0.0  0:00.00 cpuhp/0
20 root      20    0      0      0      0 S  0.0  0.0  0:00.00 cpuhp/1
21 root      rt    0      0      0      0 S  0.0  0.0  0:00.02 migration/1
22 root      20    0      0      0      0 S  0.0  0.0  0:00.02 ksoftirqd/1
24 root      0 -20    0      0      0 I  0.0  0.0  0:00.00 kworker/1:0H-xfs-log/nvme0n1p1

[root@Nikita ~]# ps -ef | grep httpd
root      2963      1  0 14:20 ?        00:00:00 /usr/sbin/httpd -DFOREGROUND
apache    2964    2963  0 14:20 ?        00:00:00 /usr/sbin/httpd -DFOREGROUND
apache    2965    2963  0 14:20 ?        00:00:00 /usr/sbin/httpd -DFOREGROUND
apache    2966    2963  0 14:20 ?        00:00:00 /usr/sbin/httpd -DFOREGROUND
apache    2986    2963  0 14:20 ?        00:00:00 /usr/sbin/httpd -DFOREGROUND
apache    3204    2963  0 14:21 ?        00:00:00 /usr/sbin/httpd -DFOREGROUND
root      3529    3380  0 14:25 pts/1    00:00:00 grep --color=auto httpd

[root@Nikita ~]# systemctl stop httpd
[root@Nikita ~]# systemctl status httpd
o httpd.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/httpd.service; enabled; preset: disabled)
   Active: inactive (dead) since Sun 2026-09-27 14:26:14 UTC; 10s ago
     Duration: 6min 642ms
    Docs: man:httpd.service(8)
   Process: 2963 ExecStart=/usr/sbin/httpd $OPTIONS -DFOREGROUND (code=exited, status=0/SUCCESS)
   Main PID: 2963 (code=exited, status=0/SUCCESS)
   Status: "Total requests: 3; Idle/Busy workers 100/0;Requests/sec: 0.00836; Bytes served/sec:  5 B/sec"
      CPU: 481ms

Sep 27 14:20:12 Nikita systemd[1]: Starting httpd.service - The Apache HTTP Server...
Sep 27 14:20:12 Nikita httpd[2963]: AH00558: httpd: Could not reliably determine the server's fully qualified domain name, using fe80::8ff:dfff:feaa:a9c-
Sep 27 14:20:12 Nikita systemd[1]: Started httpd.service - The Apache HTTP Server.
Sep 27 14:20:12 Nikita httpd[2963]: Server configured, listening on: port 80
Sep 27 14:26:13 Nikita systemd[1]: Stopping httpd.service - The Apache HTTP Server...
Sep 27 14:26:14 Nikita systemd[1]: httpd.service: Deactivated successfully.
```

```
[root@Nikita ~]# systemctl start httpd
[root@Nikita ~]# systemctl status httpd
• httpd.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/httpd.service; enabled; preset: disabled)
   Active: active (running) since Sun 2026-09-27 14:26:45 UTC; 11s ago
     Docs: man:httpd.service(8)
  Main PID: 3600 (httpd)
    Status: "Total requests: 0; Idle/Busy workers 100/0; Requests/sec: 0; Bytes served/sec: 0 B/sec"
     Tasks: 177 (limit: 1059)
    Memory: 13.9M
       CPU: 68ms
    CGroup: /system.slice/httpd.service
            └─3600 /usr/sbin/httpd -DFOREGROUND
            └─3601 /usr/sbin/httpd -DFOREGROUND
            └─3602 /usr/sbin/httpd -DFOREGROUND
            └─3645 /usr/sbin/httpd -DFOREGROUND
            └─3647 /usr/sbin/httpd -DFOREGROUND
```

```
Sep 27 14:26:45 Nikita systemd[1]: Starting httpd.service - The Apache HTTP Server...
Sep 27 14:26:45 Nikita httpd[3600]: AH00558: httpd: Could not reliably determine the server's fully qualified domain name, using fe80::8ff:dfff:feaa:a9c
Sep 27 14:26:45 Nikita systemd[1]: Started httpd.service - The Apache HTTP Server.
Sep 27 14:26:45 Nikita httpd[3600]: Server configured, listening on: port 80
```

```
[root@Nikita ~]# pgrep -o httpd
3600
[root@Nikita ~]# ps p 3600 -f
UID      PID    PPID  C  STIME TTY          STAT      TIME CMD
root      3600      1   0  14:26 ?        Ss         0:00 /usr/sbin/httpd -DFOREGROUND
[root@Nikita ~]# sudo kill 3600
[root@Nikita ~]# ps -p pid
error: process ID list syntax error
```

```
Usage:
ps [options]
```

```
For more details see ps(1).
[root@Nikita ~]# ps -p 3600
  PID TTY          TIME CMD
[root@Nikita ~]# ps -p 3600
  PID TTY          TIME CMD
[root@Nikita ~]# systemctl status httpd
o httpd.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/httpd.service; enabled; preset: disabled)
   Active: inactive (dead) since Sun 2026-09-27 14:29:28 UTC; 5min ago
 Duration: 2min 42.887s
     Docs: man:httpd.service(8)
  Process: 3600 ExecStart=/usr/sbin/httpd $OPTIONS -DFOREGROUND (code=exited, status=0/SUCCESS)
 Main PID: 3600 (code=exited, status=0/SUCCESS)
    Status: "Total requests: 0; Idle/Busy workers 100/0; Requests/sec: 0; Bytes served/sec: 0 B/sec"
       CPU: 211ms
```

```
Sep 27 14:26:45 Nikita systemd[1]: Starting httpd.service - The Apache HTTP Server...
Sep 27 14:26:45 Nikita httpd[3600]: AH00558: httpd: Could not reliably determine the server's fully qualified domain name, using fe80::8ff:dfff:feaa:a9c
Sep 27 14:26:45 Nikita systemd[1]: Started httpd.service - The Apache HTTP Server.
Sep 27 14:26:45 Nikita httpd[3600]: Server configured, listening on: port 80
Sep 27 14:29:28 Nikita systemd[1]: httpd.service: Deactivated successfully.
```

```
[root@Nikita ~]#
```

Q10. Search & Texting Process:

```
aws | [Alt+S]
[root@Nikita ~]# cd LinuxExam
[root@Nikita LinuxExam]# vim student.txt
[root@Nikita LinuxExam]# cat student.txt
1 Nikita
2 Tanuja
3 Suraj
4 Tanvi
5 Avishkar
6 Harshal sir
7 Ruchita
8 Bhakti
9 Shiva
10 Roshan
[root@Nikita LinuxExam]# grep "a" student.txt
1 Nikita
2 Tanuja
3 Suraj
4 Tanvi
5 Avishkar
6 Harshal sir
7 Ruchita
8 Bhakti
9 Shiva
10 Roshan
[root@Nikita LinuxExam]# grep -i "a" student.txt
1 Nikita
2 Tanuja
3 Suraj
4 Tanvi
5 Avishkar
6 Harshal sir
7 Ruchita
8 Bhakti
9 Shiva
10 Roshan
[root@Nikita LinuxExam]# grep -ic "a" student.txt
10
[root@Nikita LinuxExam]# sort student.txt
1 Nikita
10 Roshan
2 Tanuja
3 Suraj
4 Tanvi
5 Avishkar
6 Harshal sir
7 Ruchita
8 Bhakti
9 Shiva
[root@Nikita LinuxExam]# find . -name "*.txt"
./Backups/student_backup.txt
./linux_course.txt
./student.txt
[root@Nikita LinuxExam]#
```

Q12. Linux Networking :

```
aws [Search] [Alt+S]

[root@Nikita LinuxExam]# hostname
Nikita
[root@Nikita LinuxExam]# ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid lft forever preferred lft forever
    inet6 ::1/128 scope host noprefixroute
        valid lft forever preferred lft forever
2: ens5: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 9001 qdisc mq state UP group default qlen 1000
    link/ether 0a:ff:d1:aa:a9:cd brd ff:ff:ff:ff:ff:ff
    altname enp0s5
    altname eni-01fd9affda96566c7
    altname device-number-0.0
    inet 172.31.1.94/20 metric 512 brd 172.31.15.255 scope global dynamic ens5
        valid lft 2745sec preferred lft 2745sec
    inet6 fe80::8ff:d1ff:feaa:a9cd/64 scope link proto kernel_ll
        valid lft forever preferred_lft forever
[root@Nikita LinuxExam]# ip link
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN mode DEFAULT group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
2: ens5: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 9001 qdisc mq state UP mode DEFAULT group default qlen 1000
    link/ether 0a:ff:d1:aa:a9:cd brd ff:ff:ff:ff:ff:ff
    altname enp0s5
    altname eni-01fd9affda96566c7
    altname device-number-0.0
[root@Nikita LinuxExam]# ip route
default via 172.31.0.1 dev ens5 proto dhcp src 172.31.1.94 metric 512
172.31.0.0/20 dev ens5 proto kernel scope link src 172.31.1.94 metric 512
172.31.0.1 dev ens5 proto dhcp scope link src 172.31.1.94 metric 512
172.31.0.2 dev ens5 proto dhcp scope link src 172.31.1.94 metric 512
[root@Nikita LinuxExam]# ping -c 4 8.8.8.8
ping: invalid argument: '4.8.8.8'
[root@Nikita LinuxExam]# ping -c 4 google.com
PING google.com (142.250.207.142) 56(84) bytes of data.
64 bytes from pnbomb-bb-in-f14.1e100.net (142.250.207.142): icmp_seq=1 ttl=114 time=1.87 ms
64 bytes from pnbomb-bb-in-f14.1e100.net (142.250.207.142): icmp_seq=2 ttl=114 time=1.89 ms
64 bytes from pnbomb-bb-in-f14.1e100.net (142.250.207.142): icmp_seq=3 ttl=114 time=1.89 ms
64 bytes from pnbomb-bb-in-f14.1e100.net (142.250.207.142): icmp_seq=4 ttl=114 time=1.89 ms

--- google.com ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3004ms
rtt min/avg/max/mdev = 1.871/1.888/1.902/0.011 ms
[root@Nikita LinuxExam]# curl https://example.com
<!doctype html><html lang="en"><head><title>Example Domain</title><link rel="icon" href="data:;"/><meta name="viewport" content="width=device-width, initial-scale=1"/><style>body{background
#eee;width:60%;margin:15px auto;font-family:sans-serif;h1{font-size:1.5em}div{opacity:0.8}a:visited{color:#348}</style></head><body><div><h1>Example Domain</h1><p>This
domain is for use in documentation examples without needing permission. Avoid use in operations.</p><p><a href="https://iana.org/domains/example">Learn more</a></p></div></body></html>
[root@Nikita LinuxExam]#
```

i-0468f668416315de (Linux_exam)

```
ping: invalid argument: '4.8.8.8'
[root@Nikita LinuxExam]# ping -c 4 google.com
PING google.com (142.250.207.142) 56(84) bytes of data.
64 bytes from pnbomb-bb-in-f14.1e100.net (142.250.207.142): icmp_seq=1 ttl=114 time=1.87 ms
64 bytes from pnbomb-bb-in-f14.1e100.net (142.250.207.142): icmp_seq=2 ttl=114 time=1.90 ms
64 bytes from pnbomb-bb-in-f14.1e100.net (142.250.207.142): icmp_seq=3 ttl=114 time=1.89 ms
64 bytes from pnbomb-bb-in-f14.1e100.net (142.250.207.142): icmp_seq=4 ttl=114 time=1.89 ms

--- google.com ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3004ms
rtt min/avg/max/mdev = 1.871/1.888/1.902/0.011 ms
[root@Nikita LinuxExam]# curl https://example.com
<!doctype html><html lang="en"><head><title>Example Domain</title><link rel="icon" href="data:;"/><meta name="viewport" content="width=device-width, initial-scale=1"/><style>body{background
#eee;width:60%;margin:15px auto;font-family:sans-serif;h1{font-size:1.5em}div{opacity:0.8}a:visited{color:#348}</style></head><body><div><h1>Example Domain</h1><p>This
domain is for use in documentation examples without needing permission. Avoid use in operations.</p><p><a href="https://iana.org/domains/example">Learn more</a></p></div></body></html>
[root@Nikita LinuxExam]#
```

```
nslookup google.com
Server:      172.31.0.2
Address:     172.31.0.2#53

Non-authoritative answer:
Name:   google.com
Address: 142.251.126.102
Name:   google.com
Address: 142.251.126.101
Name:   google.com
Address: 142.251.126.100
Name:   google.com
Address: 142.251.126.113
Name:   google.com
Address: 142.251.126.138
Name:   google.com
Address: 142.251.126.139
Name:   google.com
Address: 2404:6800:4000:1015:8a
Name:   google.com
Address: 2404:6800:4000:1015:8b
Name:   google.com
Address: 2404:6800:4000:1015:71
Name:   google.com
Address: 2404:6800:4000:1015:66

[root@Nikita LinuxExam]# ss -int
State      Recv-Q      Send-Q      Local Address:Port      Peer Address:Port
ESTAB      0            0            172.31.1.94:22          13.233.177.5:56618
cubic wscale:7,7 rto:210 rtt:0.945/0.052 ato:60 mss:1448 pmtu:9001 rcvms:688 advms:8949 cwnd:10 ssthresh:34 bytes_sent:245633 bytes_acked:245633 bytes_received:59004 segs_out:1
862 segs_in:2971 data_segs_out:1805 data_segs_in:1438 send:122582011bps lastsnd:20 lastrcv:20 pacing_rate:245066768bps delivery_rate:189694320bps delivered:1806 app_limited bus
y:3350ms rcv_space:56587 rcv_ssthresh:63562 minrtt:0.88 snd_wnd:56704 rcv_wnd:63616
[root@Nikita LinuxExam]#
```

Challenge:

```
root@Nikita LinuxExam)# systemctl status httpd
b httpd.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/httpd.service; enabled; preset: disabled)
   Active: inactive (dead) since Sun 2026-09-27 14:29:28 UTC; 32min ago
 Duration: 2min 42.887s
    Docs: man:httpd.service(8)
  Process: 3600 ExecStart=/usr/sbin/httpd $OPTIONS -DFOREGROUND (code=exited, status=0/SUCCESS)
 Main PID: 3600 (code=exited, status=0/SUCCESS)
   Status: "Total requests: 0; Idle/Busy workers 100/0;Requests/sec: 0; Bytes served/sec: 0 B/sec"
    CPU: 211ms

Sep 27 14:26:45 Nikita systemd[1]: Starting httpd.service - The Apache HTTP Server...
Sep 27 14:26:45 Nikita httpd[3600]: AH00558: httpd: Could not reliably determine the server's fully qualified domain name, using fe80::8ff:diff:feaa:a9c
Sep 27 14:26:45 Nikita systemd[1]: Started httpd.service - The Apache HTTP Server.
Sep 27 14:26:45 Nikita httpd[3600]: Server configured, listening on: port 80
Sep 27 14:29:28 Nikita systemd[1]: httpd.service: Deactivated successfully.

root@Nikita LinuxExam)# systemctl start httpd
root@Nikita LinuxExam)# systemctl status httpd
• httpd.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/httpd.service; enabled; preset: disabled)
   Active: active (running) since Sun 2026-09-27 15:02:34 UTC; 3s ago
    Docs: man:httpd.service(8)
 Main PID: 5023 (httpd)
   Status: "Started, listening on: port 80"
    Tasks: 177 (limit: 1059)
  Memory: 13.7M
     CPU: 78ms
    CGroup: /system.slice/httpd.service
            └─5023 /usr/sbin/httpd -DFOREGROUND
              └─5053 /usr/sbin/httpd -DFOREGROUND
                └─5054 /usr/sbin/httpd -DFOREGROUND
                  └─5075 /usr/sbin/httpd -DFOREGROUND
                    └─5126 /usr/sbin/httpd -DFOREGROUND

Sep 27 15:02:34 Nikita systemd[1]: Starting httpd.service - The Apache HTTP Server...

Sep 27 15:02:34 Nikita systemd[1]: Starting httpd.service - The Apache HTTP Server...
Sep 27 15:02:34 Nikita httpd[5023]: AH00558: httpd: Could not reliably determine the server's fully qualified domain name, using fe80::8ff:diff:feaa:a9c
Sep 27 15:02:34 Nikita systemd[1]: Started httpd.service - The Apache HTTP Server.
Sep 27 15:02:34 Nikita httpd[5023]: Server configured, listening on: port 80

[root@Nikita LinuxExam]# systemctl enable httpd
[root@Nikita LinuxExam]# ss -ltnp | grep :80
[root@Nikita LinuxExam]# grep Listen /etc/httpd/conf/httpd.conf
# Listen: Allows you to bind Apache to specific IP addresses and/or
# Change this to Listen on a specific IP address, but note that if
# Listen 12.34.56.78:80
Listen 80

[root@Nikita LinuxExam]# ss -ltnp | grep :80
LISTEN 0      511          *:*          *:*        users:((("httpd",pid=5126,fd=4),("httpd",pid=5075,fd=4),("httpd",pid=5054,fd=4),("httpd",pid=5023,fd=4)))
[root@Nikita LinuxExam]# █

I-0468f6668416315de (Linux_exam)
```